

Hello everyone, I'm a technical support engineer from D-Robotics. This article aims to help developers solve the following common issues:

- System fails to start after flashing
- Wi-Fi and other modules are abnormal (unavailable) after system startup
- Some software functions cannot load properly

Problem Background

During development and usage, everyone uses different SD card brands and models. These SD cards have differences in **power-on timing**, **boot methods**, and other aspects, causing system compatibility issues on certain cards.

To solve this problem, we have upgraded the **Miniboot** bootloader, thereby **significantly improving compatibility with mainstream SD cards**.

Miniboot Introduction

Miniboot is a lightweight bootloader that mainly performs initialization during early system startup.

In the RDK X5 platform, Miniboot is embedded in the onboard **NAND Flash** and is responsible for loading the file system and core components from the SD card after startup.

Miniboot Upgrade Guide

We provide two upgrade paths for users in different situations:



Option 1: System can start normally — Automatic online

upgrade

Applicable when the system can be accessed and the network is working normally.

Simply execute the following command for one-click upgrade:

bash

Copy and edit

```
root@ubuntu:~# rdk-miniboot-update
```



Option 2: System cannot start — Manual flashing of Miniboot

Applicable when the system is abnormal or fails to start.

1. Environment Preparation

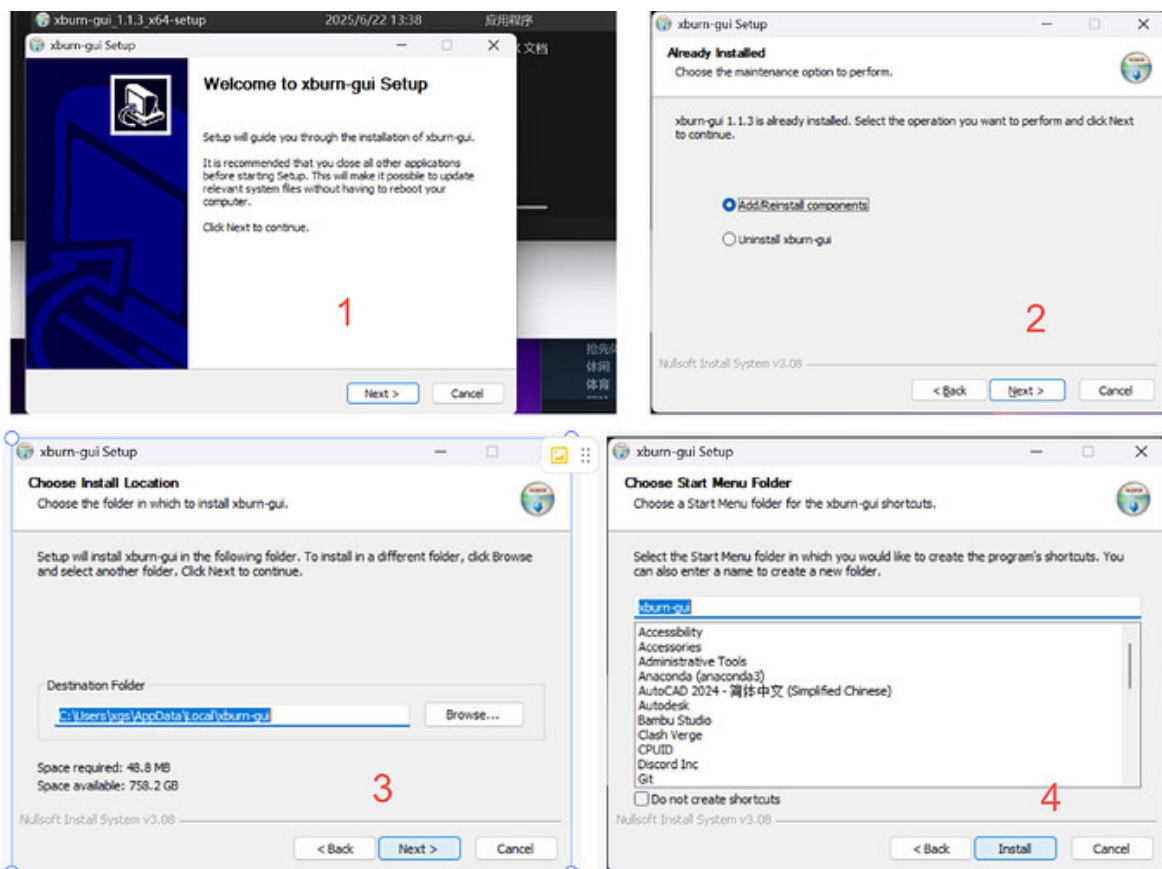
Tool Category	Download Link	Description
Flashing Tool	http://rdk-doc.oss-cn-beijing.aliyuncs.com/doc/assets/rdk_x5/burn/xburn-gui_1.1.3_x64-setup.exe	Tool for flashing miniboot system (includes various drivers for RDK series)
miniboot	Recommended version: https://archive.d-robotics.cc/downloads/miniboot/rdk_x5/product_20250610.zip All versions: Index of /downloads/miniboot/rdk_x5/	Lightweight boot bootloader
Connection Cable	Micro USB cable	

2. Flashing Tool Installation

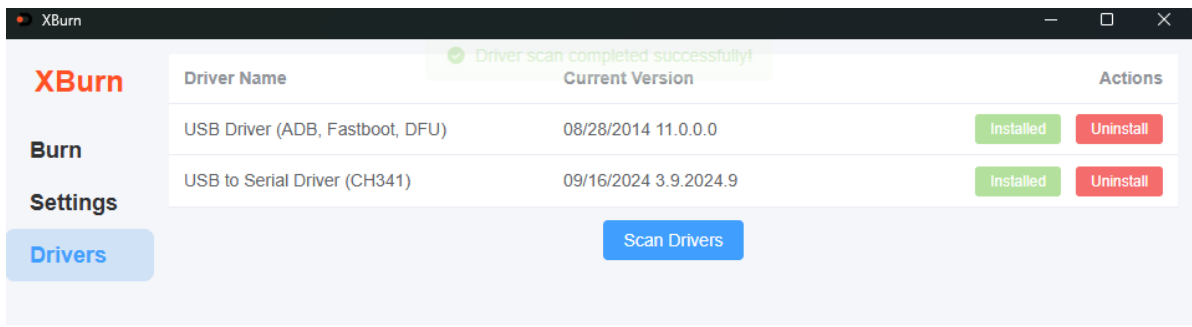
1. Download and install the flashing tool (must be operated on Windows system)



We will provide macOS and Ubuntu platform support in the future



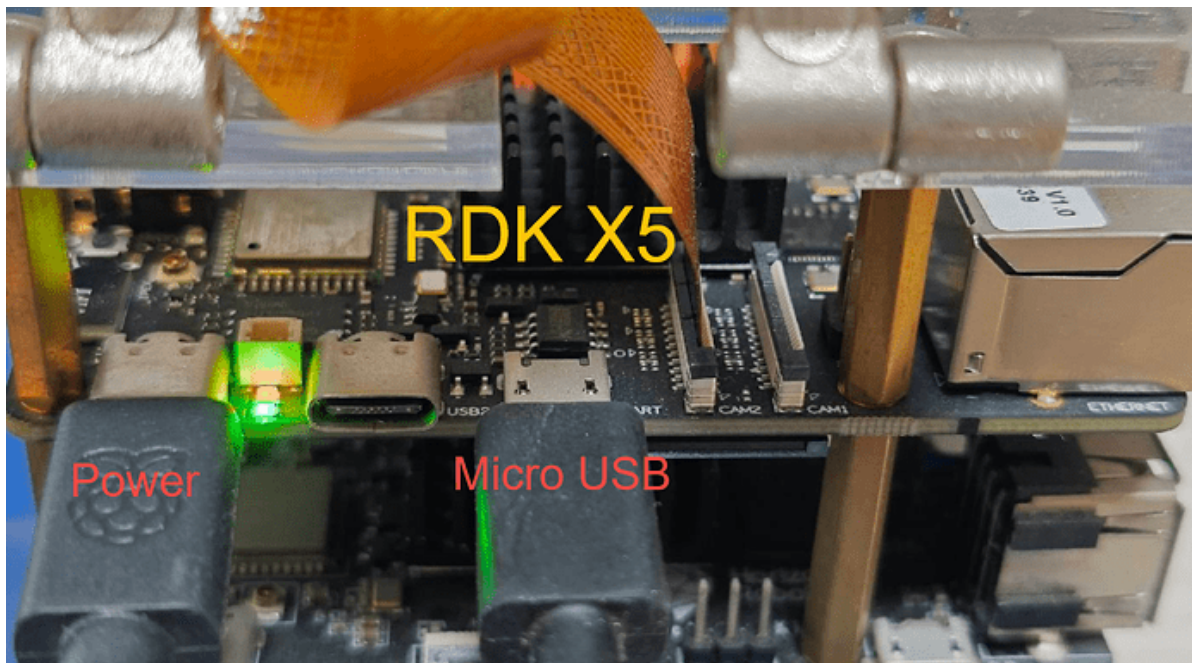
1. Install drivers: Ensure relevant USB drivers are installed



3. Hardware Connection

Use Micro-USB data cable to connect the development board and PC. Note:

- The Micro-USB port on RDK X5 is also the **Debug UART interface**
- Ensure stable connection, recommend using high-quality data cable



4. Software Configuration Parameters

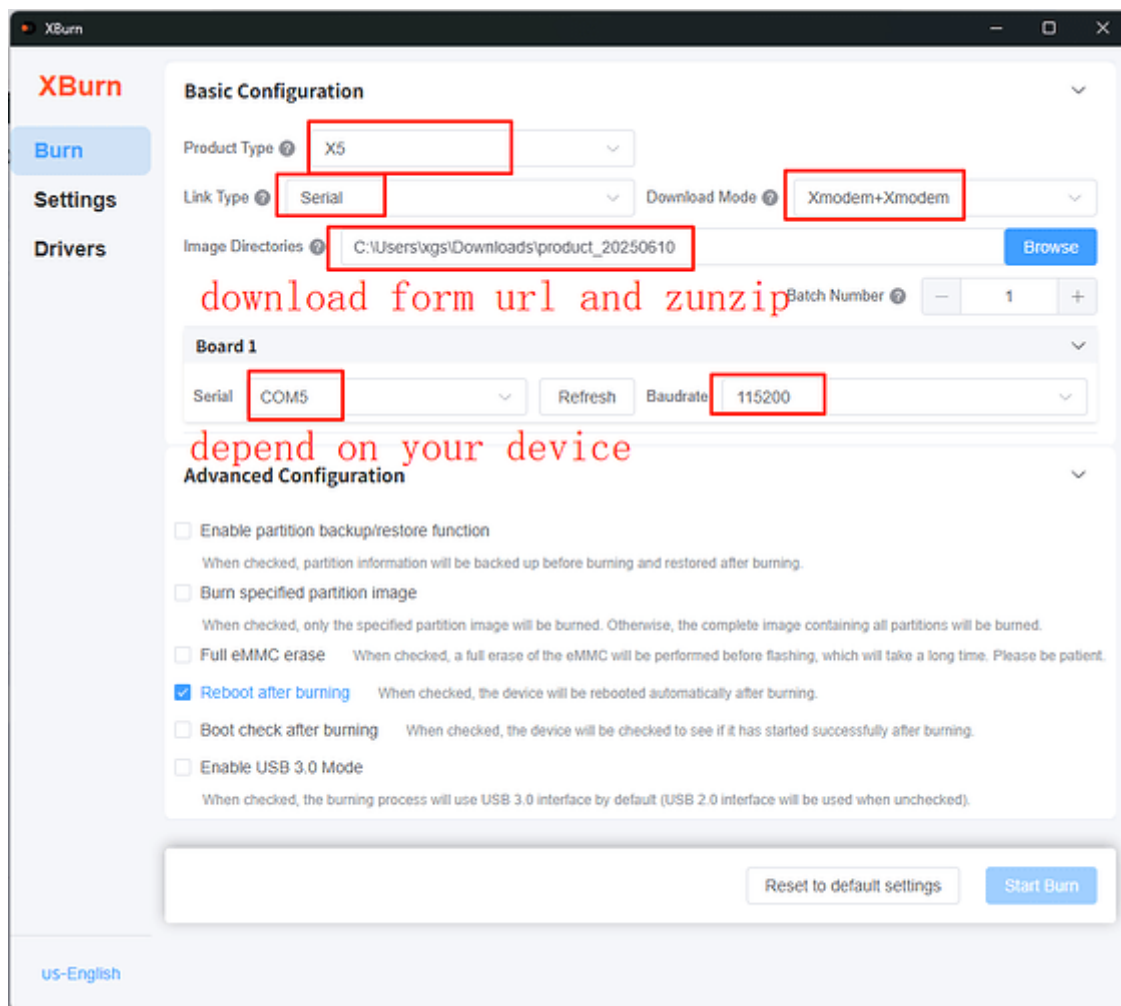
Open the flashing tool and set parameters as follows:

wrap:normal;}

Parameter Item	Configuration Content
Product Type	X5
Link Type	Serial
Download Mode	Xmodem+Xmodem
Serial	Select actual serial port
Baudrate	115200

Configure as shown in the figure below:

Refer to the figure below for specific configuration



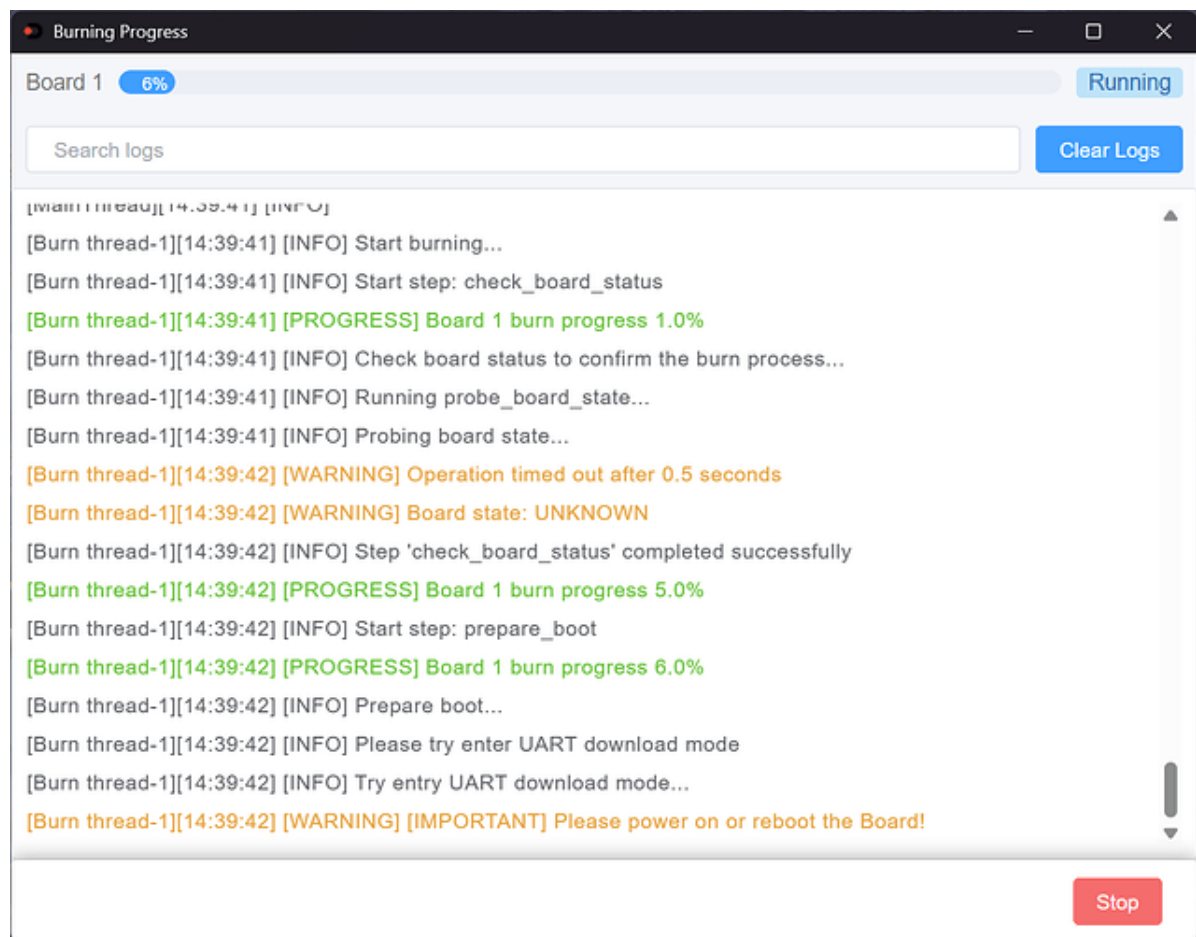
5. Start Flashing Process



Power on and click "Flash"

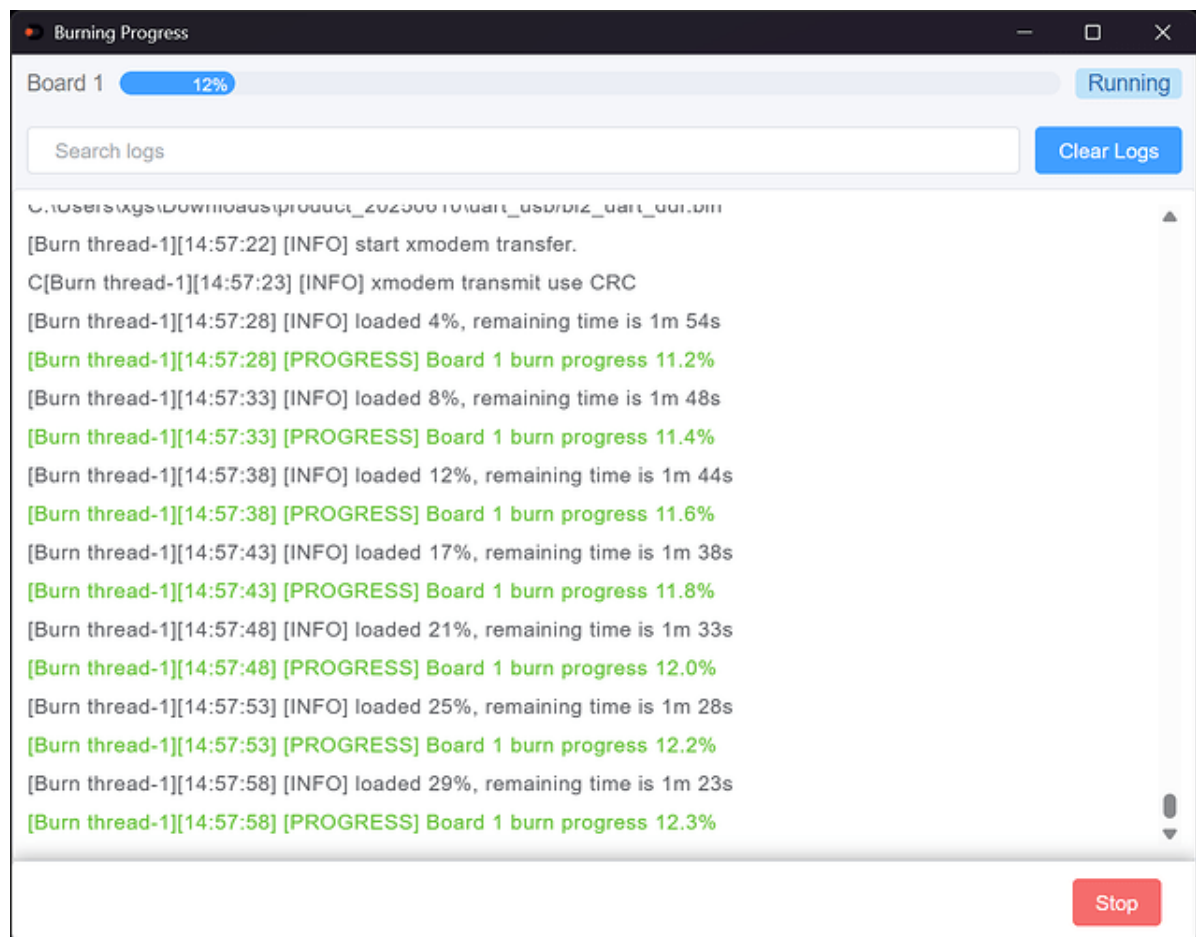
The software will progress to 6% and then pause, prompting to wait for restart.

![Flashing start interface]



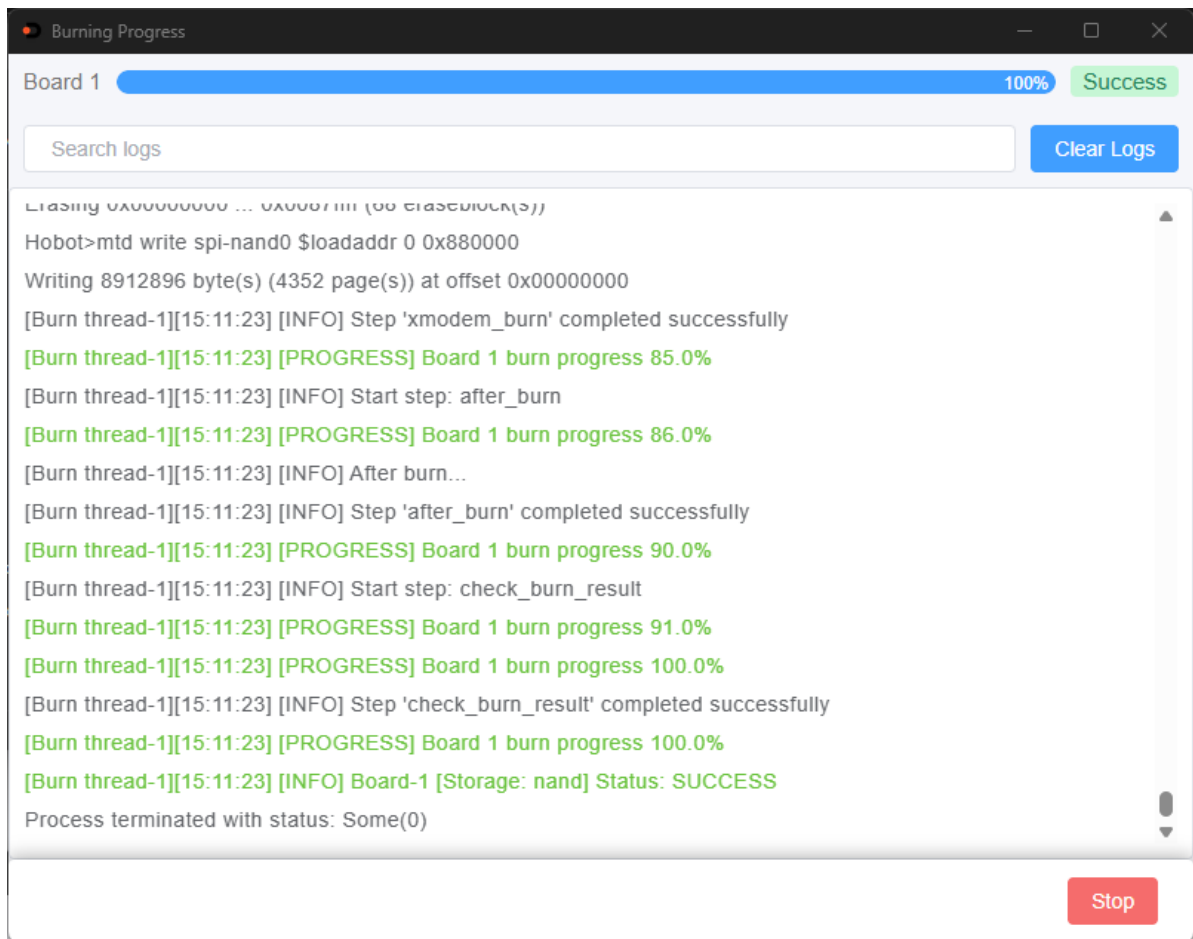
Power off → Power on again

This officially enters the flashing phase



Flashing Complete

The flashing process takes about 10 minutes (serial port speed is slow, please wait patiently).



Verify Miniboot Update Success

After system startup, execute the following command to view current system information:

```
root@ubuntu:~# rdkos_info
```

This command will output key parameters such as current operating system version, Miniboot version, etc.


```
cpu5: 300      1200    1500
cpu6: 300      1200    1500
cpu7: 300      1200    1500
bpu status information---->
      min(M)    cur(M)  max(M)  ratio
bpu0: 500      1000    1000    0
ddr frequency information---->
      min(M)    cur(M)  max(M)
ddr:  266      4266    4266
GPU gc8000 frequency information---->
      min(M)    cur(M)  max(M)
gc8000: 200    1000    1000

[Total Memory]:      3.0Gi
[Used Memory]:       518Mi
[Free Memory]:       1.9Gi
[ION Memory Size]:    128MB
[ION Reserved Memory Size]: 320MB
[ION Carveout Memory Size]: 320MB

[RDK OS Version]:
  3.2.3

[RDK Kernel Version]:
  Linux ubuntu 6.1.83 #1 SMP PREEMPT Thu May 15 14:54:05 CST 2025 aarch64 aarch64 aarch64 GN
  U/Linux

[RDK Miniboot Version]:
  U-Boot 2022.10+ (Jun 10 2025 - 19:30:58 +0800)

[RDK Packages List]:
  hobot-audio-config/unknown,now 3.0.1-20250515180311 arm64 [installed]
  hobot-boot/unknown,now 3.0.2-20250515180311 arm64 [installed]
  hobot-camera/unknown,now 3.0.1-20250515180311 arm64 [installed]
  hobot-configs/unknown,now 3.0.1-20250515180311 arm64 [installed]
  hobot-display/unknown,now 3.0.0-20241018162911 arm64 [installed]
  hobot-dnn/unknown,now 3.0.1-20250110154647 arm64 [installed]
  hobot-dtb/unknown,now 3.0.2-20250515180311 arm64 [installed]
  hobot-io-samples/unknown,now 3.0.0-20250115150941 arm64 [installed]
  hobot-io/unknown,now 3.0.1-20250516160512 arm64 [installed]
  hobot-kernel-headers/unknown,now 3.0.1-20250515180311 arm64 [installed]
  hobot-miniboot/unknown,now 3.0.1-20250515180311 arm64 [installed]
```